
Building a Profile Hidden Markov Model for the detection of Kunitz Type Protease Inhibitor Domain

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Abstract

The Kunitz domain is a specific, highly conserved structural motif found in various proteins. The domains are relatively small typically consisting of about 50 to 60 amino acids, the structure is locked into place by three disulfide bonds. Thanks to its small size, structural conservation and biological relevance the Kunitz domain is an extensively studied model structure key target in protein annotation and therapeutic development. This study focuses on building a developed structure-informed Hidden Markov Model (HMM) to identify Kunitz inhibitory domain in protein sequences. The profile HMM was built using high-resolution structural data from PDB and trained on a multiple sequence alignment. The model was evaluated on curated positive and negative datasets, functioning as a binary classifier. The model performance was evaluated using Matthews Correlation Coefficient (MCC), confirming its reliability for detecting Kunitz domains.

1 Introduction

1.1 The Kunitz Domain

Kunitz type proteins identified with the Pfam entry PF00014 on Uniprot are an important group of ubiquitous protease inhibitors found spanning the evolutionary tree from microbes to mammals. Kunitz domains are the active part of these proteins that inhibit the function of protein degrading enzymes. Domains of Kunitz-type are protease inhibitors, comprising a chain of around 60 amino acid residues stabilized by three disulphide bonds, created by 6 highly conserved cysteine. These disulfide bridges play a dual role in stabilizing the native conformation of the domain and facilitating its interaction with protease. The domains are highly resistant to heat, extreme pH levels, and degradation; in vertebrates Kunitz inhibitors play a major role in inflammatory processes while in invertebrates they are involved in a range of diverse functional roles (Ranasinghe & McManus, 2013). Examples of Kunitz-type protease inhibitors are aprotinin and tissue factor pathway inhibitor (TFPI) that have a crucial role in regulating blood clotting (St Charles et al., 2000). The level of conservation through species and the functional biological relevance highlight the interest in the development of computational tools for their detection.

1.2 Hidden Markov Models (HMMs)

Hidden Markov Models (HMMs) are powerful statistical tools widely used in computational biology to model and analyze biological sequences such as proteins and nucleic acids (Finn et al., 2011). It is a statistical model

used to describe a system that transitions through a sequence of states over time, where the actual states are hidden from you, but they generate output that you can observe. HMMs are incredibly powerful for interpreting sequences of data where there is underlying uncertainty. It's composed by: states, initial probability, transition probabilities and emission probabilities. Together, these generate a probabilistic framework that can capture both conservation and variability across related biological sequences. In protein domain analysis, HMMs are particularly effective because they can model position-specific features, including conserved residues, insertion-deletion patterns and evolutionary variability. This is achieved by training an HMM on a multiple sequence/structure alignment of known domain instances, yielding what is called a profile HMM. This profile provides a formal probabilistic model that is robust and flexible in detecting remote homologs, even in the presence of noise or low sequence identity. For this reason, HMMs have become a standard tool in sequence annotation, domain detection, and protein family classification, and are used extensively in databases such as Pfam, SMART, and InterPro.

1.3 Aims of the project

The goal of this project is to create a bioinformatics pipeline to synthesize and evaluate an HMM profile for detecting the Kunitz domain. The model is trained on structurally aligned sequences, tested on filtered datasets, and finally used to analytically determine the tolerance threshold (E-value) to maximize the accuracy, without losing information.

2 Materials & Methods

2.1 Dataset collection and filtering

Structural data for the Kunitz domain were retrieved from the RCSB Protein Data Bank (PDB) using its advanced search interface, applying the following filters:

- Pfam accession = PF00014
- X-ray resolution ≤ 3.5 Å
- Sequence length: $40 < length < 80$

The resulting custom report (412 chain entries) included for each entry the PDB identifier, the amino acid sequence and the chain identifier. The raw table was then cleaned filtered and processed using a custom Python script that in three steps removed entirely empty rows, chains without an associated sequence and sequences shorter than 40 or longer than 80 residues. This filter yielded a final set of 125 sequences, used as input for the redundancy-reduction and structural-alignment step described below.

2.2 PDBeFold & MMSeqs2

The 125 filtered sequences were exported to FASTA format, with each header encoding the PDB identifier and chain. To avoid training the HMM on near-duplicate structures, the retrieved data was processed in order to reduce redundancy with MMSeqs2 (Steinberger & Söding, 2017). MMSeqs2 created different clusters and only one representative sequence per cluster was retained. The cluster representatives were then submitted to PDBeFold (Krissinel & Henrick, 2004), an online structural-alignment server, to generate a structure-based multiple sequence alignment sharing more than 90% of sequence identity. Structural, rather than purely sequence-based, alignment was chosen because Kunitz domains can be remotely related in sequence while retaining a highly conserved three-dimensional fold; aligning on structure therefore captures the conserved scaffold more reliably than a standard sequence alignment would. The alignment returned by PDBeFold was post-processed and the sequences with a RMSD (Root Mean Square Deviation) bigger than 1 were manually removed, producing the final cleaned alignment used for HMMtraining.

Structure	RMSD	Q-score
1AAP:B	0.4434	0.6641
1F5R:I	0.4283	0.6533
1KNT:A	0.6774	0.6574
1TFX:D	0.4890	0.6382
1YC0:I	0.3782	0.5668
1YLD:B	0.4181	0.6656
1ZR0:D	0.3713	0.5941
3FP7:J	0.4679	0.8627
3L33:H	0.5540	0.7067
4DTG:K	0.4534	0.6192
4U30:W	0.5958	0.6770
4U32:X	0.3099	0.6963
5JB6:C	0.4911	0.6381
8PTI:A	1.2675	0.5559

Table 1: Structural alignment results. Q-score and RMSD (Å) for the 14 representative PDB structures used to build the Kunitz-domain multiple structural alignment. Thirteen structures cluster tightly (RMSD 0.3–0.7 Å), while 8pti:A is a clear outlier with an RMSD of 1.27 Å. This structure was manually excluded from the alignment prior to HMM training, since a poor structural superposition would otherwise introduce noise into the profile.

2.3 HMM building & HMM search

A profile HMM was calculated from the cleaned structural alignment using hmmbuild (HMMER v3) which estimates model parameters from the MSA (Figure 1). Two evaluation sets were then retrieved from UniProt/Swiss-Prot (reviewed entries only): a positive set, comprising all reviewed proteins annotated with the Kunitz Pfam domain (PF00014, 398 sequences), and a negative set comprising all reviewed proteins without that annotation. In order to not produce data leakage the PDB identifiers used for training were mapped back to their corresponding UniProt accessions via the SIFTS PDB-to-UniProt cross-reference table (Dana et al., 2019). This identified 8 UniProt entries corresponding to the 14 PDB IDs of the dataset. The 8 UniProt accessions already used for training were removed from the positive set, leaving 390 positive sequences. Both sets were then searched against the trained HMM profile using hmmsearch function, run with the max option (to disable heuristic filters and maximize sensitivity, which matters especially for the negative control) and a fixed database size (-Z 1000) to keep E-values comparable across datasets of different dimensions.

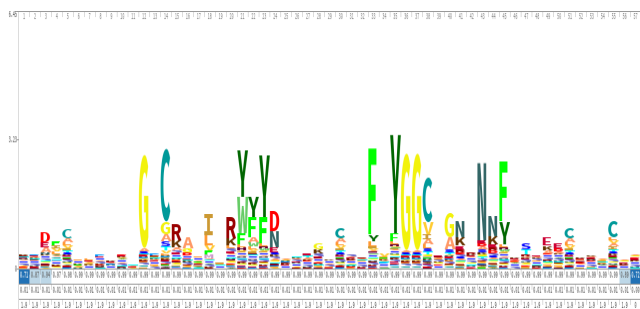


Figure 1: HMM sequence logo generated with Skyline of the trained Kunitz-domain HMM profile, covering all 57 match states. Letter height represents the frequency of the amino acids at each position. This reflects the level of conservation of the important residues throughout the evolution. In particular the tall, isolated peaks corresponding to the six cysteine residues mark the three conserved disulfide bonds that define the Kunitz fold.

2.4 Data preparation

For each hit reported by `hmmsearch`, the sequence identifier and best-domain E-value were extracted and paired with the label 1 for the positive/Kunitz set, 0 for the negative set. Because `hmmsearch` only reports sequences that scored at least one hit, negative sequences with no hit at all were recovered separately and assigned a large arbitrary E value (100) to mark them as clear non-matches. All labeled negative examples (matched and non-matched) were pooled and randomly shuffled, then split, together with the positive examples, into two equally sized folds `set_1` and `set_2`: 194/195 positives and 287,553/287,552 negatives each.

2.5 Performance evaluation

Model validation was implemented using a Python script called `performance.py` that used the E-value returned by `hmmsearch` as a binary classifier: positive if its best-domain E-value was below a given threshold and negative if the value surpass this threshold. A range of thresholds from $1e-1$ to $1e-15$ was tested and for each one was calculated the numbers of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) in order to build a confusion matrix. Classification performance was measured by calculating Accuracy (Q2) and the Matthews Correlation Coefficient (MCC)(Figure 2). MCC was chosen as the primary criterion for selecting the optimal threshold because, unlike accuracy, it remains informative under the strong class imbalance of this dataset.

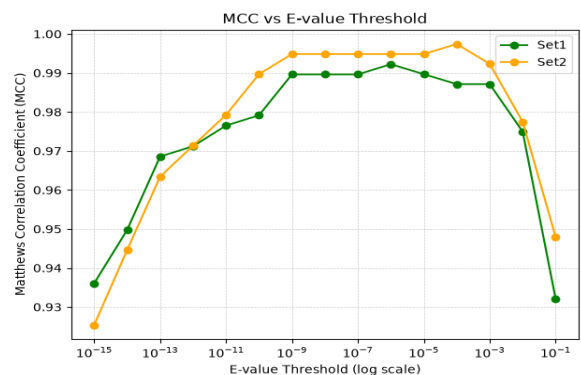


Figure 2: Matthews Correlation Coefficient as a function of the E-value classification threshold, computed independently on Set1 (green) and Set2 (orange). Both curves peak above $MCC = 0.98$ across a broad threshold range from $1e-9$ to $1e-3$, and degrade symmetrically toward both extremes. Set1 peaks at $1e-6$ ($MCC = 0.992$) while Set2 peaks slightly later, around $1e-4$ ($MCC = 0.997$).

3 Results

3.1 Training sequences selection and HMM building

Of the 14 representative structures retained after MM-seq2 clustering, 13 showed low and tightly clustered RMSD values (0.31 – 0.68 Å) and Q-scores between 0.56 and 0.86 , indicating a highly consistent structural superposition (Table 1). One structure, 8PTI:A, was a clear outlier ($RMSD = 1.27$ Å) and was therefore manually excluded from the alignment before HMM training, since including a poorly superposed structure would have introduced noise into the profile. The resulting profile HMM spans 57 match states. The sequence logo (Figure 1) highlights several highly conserved positions, most notably the six cysteine residues forming the three disulfide bonds (C1-C6, C2-C4, C3-C5) that define the Kunitz fold. This confirms the good quality of the alignment and the HMM profile built.

3.2 Model evaluation

Model performance was assessed across the 15 E-value thresholds tested independently on Set1 and Set2 (Figure 2). Both folds achieved MCC values above 0.98 over a broad threshold range from $1e-9$ to $1e-3$, but did not share a single, jointly optimal threshold: Set1 peaked at $1e-6$ ($MCC = 0.992$), while Set2 peaked at $1e-4$ ($MCC = 0.997$). Confirming that the model generalizes reliably regardless of which fold is used to fix the decision threshold.

To evaluate the discriminative power across varying thresholds, the True Positive Rate TPR (sensitivity) and False Positive Rate FPR were extrapolated for each E-value cutoff from the confusion matrix (Figure 3). At the optimal Matthews Correlation Coefficient (MCC)

thresholds (10^{-6} for Set 1 and 10^{-4} for Set 2), the model maintained a high sensitivity (TPR > 0.98), successfully identifying virtually all true Kunitz domains. Concurrently, the false positive rate remained exceptionally low (FPR < 10^{-5}), confirming that the HMM profile reliably distinguishes true domain matches from the background of more than 570,000 non-Kunitz sequences.

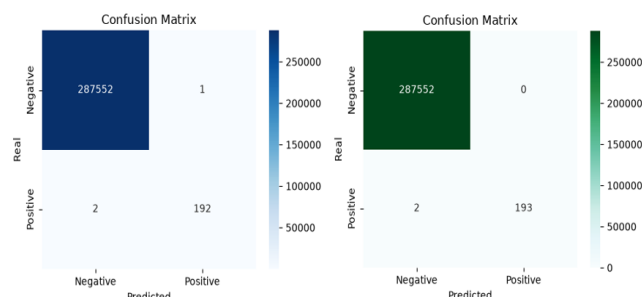


Figure 3: Confusion matrix for the HMM classifier on Set1 (blue) and Set2 (green) at the selected best E-value threshold ($1e-6$). Rows represent the true label, columns the predicted label. Out of 570,000 evaluated sequences the model produced minimal misclassifications (Set 1: 1 FP, 2 FN; Set 2: 0 FP, 1 FN), correctly identifying over 98% of true Kunitz domains while maintaining near-zero false-positive rates..

4 Conclusion

This project developed and validated a structure-informed profile HMM for the detection of the Kunitz-type protease inhibitor domain. Starting from 125 PDB structures annotated with the Kunitz Pfam domain (PF00014), redundancy reduction and RMSD-based filtering yielded 13 structurally aligned templates to build the profile. Evaluated on a strictly held-out UniProt/Swiss-Prot dataset (390 positive and over 570,000 negative sequences), the model achieved a cross-validated MCC of 0.991 at the optimal E-value threshold (10^{-6}). These results demonstrate that the profile HMM is a highly reliable and robust classifier for Kunitz domains, delivering high sensitivity with negligible false positive rates.

References

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Supplementary Material

The notebook with all the scripts and data used for this project, including the report source and the results files are available in the following dedicated repository:

https://github.com/TaiNixon/Kunitz_HMM_project